K-MEANS VS DB-SCAN

MUHAMMAD HASSAN KHALID

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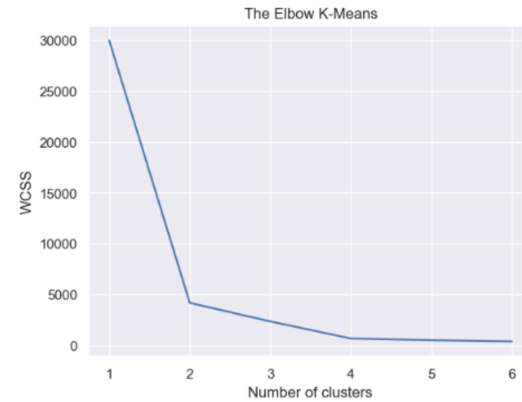
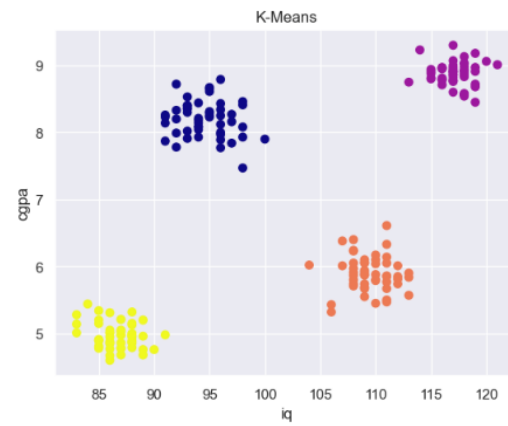
Source of Dataset:
https://www.kaggle.com/datasets/deepanshsaxena1/student-clusteringg

I am using a student_clustering.csv dataset for today's implementation of task. This dataset consists of dimensions **cgpa** and **iq** and I am using them both for my unsupervised learning. The dataset consists of 200 entries of students their **cgpa** and their respective **iq**.



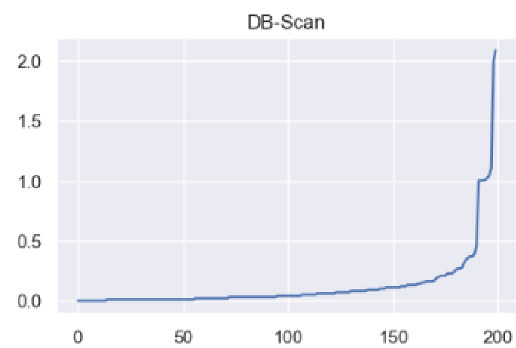


K-Means Parameters:

Number of clusters = 4

Range of clusters = 1-7

Name of the density-based algorithm that I have chosen for today's implementation of task is **DBSCAN**.



DBSCAN Parameters:

Minimum number of Points = 4

Epsilon (Eps) = 1

Comparison of K-Means and DBSCAN:

The two algorithms performed in a different way because in k-means, an object is generated to the nearest center while in DBSCAN a density-based cluster is a set of density-connected objects that is maximal regarding density-reachability. Each object not contained in some cluster is considered to be noise. **I will choose k-means for my dataset as better clusters are formed in k-mean and it is easy to understand furthermore the elbow shows a precise number of clusters to choose from. Also, in DBSCAN there is quite a noticeable noise, and the clusters are not handled well**.